

5	(a) (i)	1. wavelength: minimum distance between two points moving in phase OR distance between neighbouring or consecutive peaks or troughs OR wavelength is the distance moved by a wavefront in time T or one oscillation/cycle or period (of source)	B1	[1]
		2. frequency: number of wavefronts / (unit) time OR number of oscillations per unit time or oscillations/time	B1	[1]
	(ii)	speed = <u>distance</u> / time = <u>wavelength / time period</u> $= \lambda / T = \lambda f$	M1 A0	[1]
	(b) (i)	amplitude = 4.0 mm (allow 1 s.f.)	A1	[1]
	(ii)	wavelength = $18 / 3.75$ (= 4.8) speed = $2.5 \times 4.8 \times 10^{-2} = 12 \times 10^{-2} \text{ m s}^{-1}$ unit consistent with numerical answer, e.g. in cm s^{-1} if cm used for λ and unit changed on answer line [if $18 \text{ cm} = 3.5\lambda$ used giving speed 13 (12.9) cm s^{-1} allow max. 1].	C1 A1	[2]
	(iii)	180° or π rad	A1	[1]
	(c)	light and screen and correct positions above and below ripple tank strobe or video camera	B1 B1	[2]